



OpenAgenet/OAN

Open Governance Infrastructure for Trusted Agent Interconnection

Identity, Governance, Registration, Distribution, and Trusted Discovery

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<https://github.com/OpenAgenet>

Agenda

1. Why agent ecosystems need a trust layer
2. OAN architecture and role separation
3. Open governance and service-node authorization
4. `did: oan` resource identity model
5. Registration, distribution, and trusted discovery
6. Interoperability with MCP, A2A, ANP, and APIs
7. Engineering implementation and open materials
8. Benchmark baselines and current performance
9. Participation path and roadmap

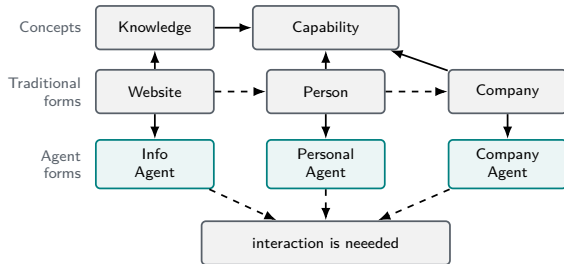
Three Takeaways

Open governance, unified resource identity, and trusted discovery are the core of OAN.

The Inevitability of Multi-Agent Collaboration: Two Perspectives

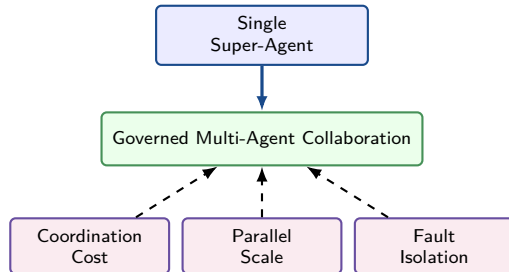
1. Macro Perspective

- Knowledge, tools, data, and responsibilities are distributed.
- Once actors become agents, collaboration becomes the default form.



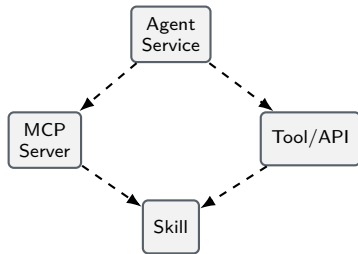
2. Engineering Perspective

- A single super-agent hits coordination, scale, and trust-domain limits.
- A governed network is more modular, scalable, and resilient.

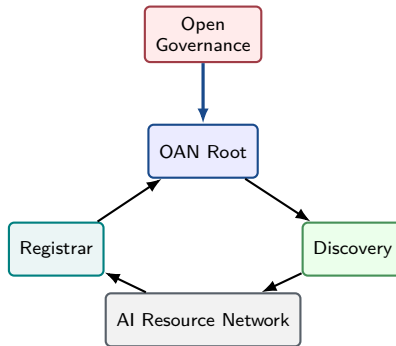


OAN implication: cross-boundary collaboration needs open identity, governance, registration, distribution, and trusted discovery.

The Agent Ecosystem Is Becoming a Network



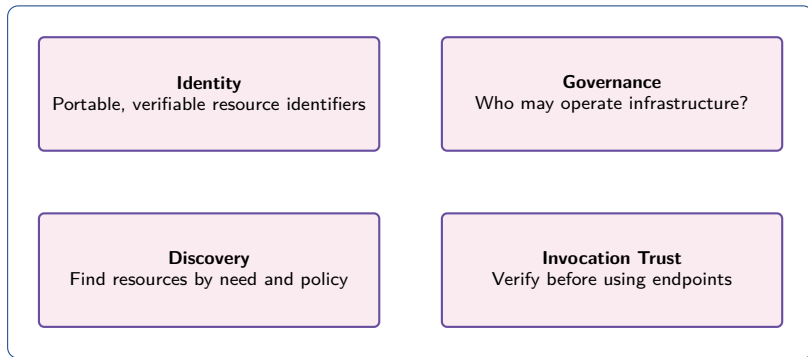
Fragmented resources and protocols



Key Message

Agent interconnection needs a trust infrastructure: identity, governance, registration, distribution, and discovery.

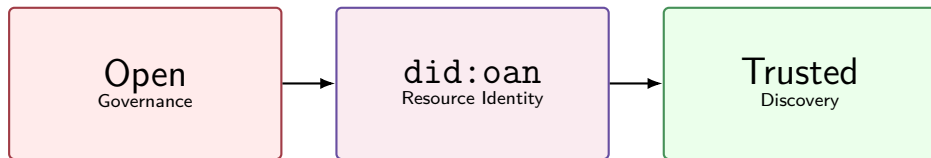
Four Missing Layers



- Protocols can connect endpoints, but ecosystems need trust, lifecycle, and evidence.
- OAN targets the missing trust and governance layer for open agent resources.

Source: White paper: Identity/Governance/Discovery/Invocation gaps; identity trust layering

OAN's Core Answer

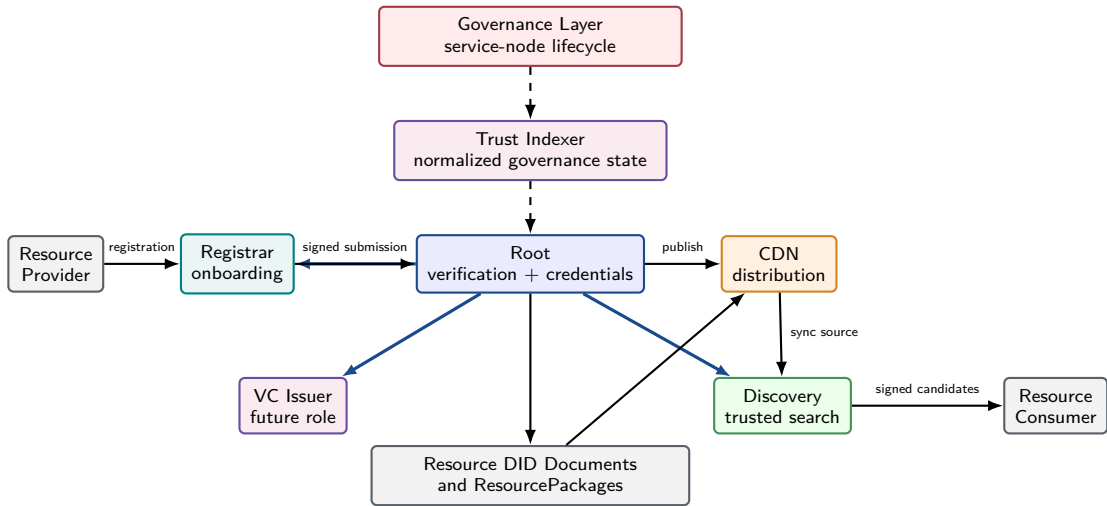


Positioning

OAN is not just a registry. It combines open governance, unified resource identity, and discovery with verifiable trust evidence.

Source: White paper: Vision and Scope; yellow paper: Protocol Stack and Trust Layers

OAN Architecture

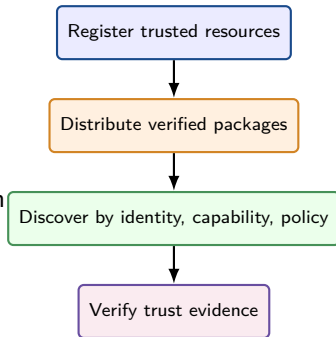


Source:

System design: Roles/Trust Flow; on-chain bulletin; trust indexer design

What OAN Is

- A trust-governed resource identity and discovery infrastructure.
- A registration, publication, distribution, discovery, and verification path for AI-agent-related resources.
- A common layer above heterogeneous interaction protocols.



Source: White paper: What OAN Is; product model: Core Positioning

What OAN Is Not

Boundary	Meaning
Not an MCP/A2A/ANP replacement	OAN complements interaction protocols with identity, governance, and discovery.
Not a generic marketplace	OAN focuses on trust-governed resource registration and discovery.
Not a universal safety solution	OAN does not guarantee model output quality or business truthfulness.
Not artifact hosting by default	Skill and API artifacts can be externally hosted and referenced with hashes.
Not a single-vendor root model	Governance state is separated from Root-issued operational credentials.

Source: White paper: What OAN Is Not; on-chain bulletin Non-goals; product model boundaries

Why the Architecture Is Valuable

Technical Advantages

- Role separation: Root, Registrar, CDN, Discovery, VC Issuer.
- Governance-state visibility for infrastructure lifecycle.
- Root-issued authorization VC for operational trust.
- DID-based portability across protocols.

Source: White paper: Design Principles; yellow paper: Security Properties

Ecosystem Advantages

- Participants can publish resources.
- Qualified operators can apply to run service nodes.
- Discovery can become domain-specific.
- Specs and docs support public review.

Positioning Against Adjacent Systems

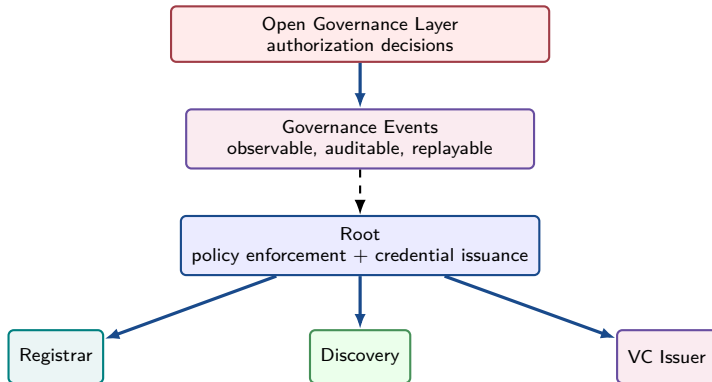
Approach	Primary Role	Trust/Governance Depth	OAN Complement
MCP	Tool/context access protocol	Usually local or platform-scoped	Makes MCP servers discoverable and verifiable as governed resources.
A2A	Agent-to-agent interaction	Protocol-level interaction focus	Adds identity, registration, and trusted discovery around services.
ANP / Agent DID	Agent identity and network interaction	Identity-oriented, variable governance	Extends toward infrastructure authorization and resource lifecycle.
Registry / Directory	Listing and lookup	Often weak governance	Adds Root verification, package proof, and signed Discovery response.
OAN	Trust-governed resource infrastructure	Identity + governance + discovery	Provides the common trust layer above heterogeneous protocols.

Interpretation

OAN is a governance and trust layer above adjacent protocols and registries, so existing systems can be onboarded instead of replaced.

Source: White paper: Relationship with MCP/A2A/ANP; yellow paper: Compatibility boundaries

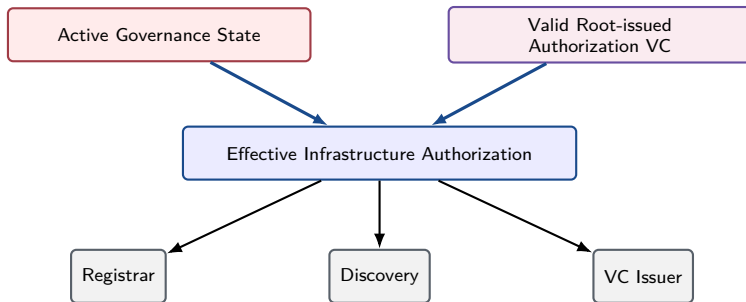
Who Controls the Root?



Message

Root is not a closed black-box authority. It enforces governance-backed trust decisions.

Effective Service-Node Authorization



Effective Authorization = Active Governance State + Valid Root-issued VC

Source: On-chain bulletin: Architectural Position; identity trust layering

Authorization Decision Matrix

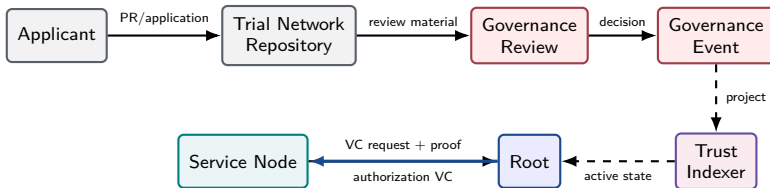
Governance State	Root VC	Service Result	Why
Active	Valid	Accept	The node is currently authorized and has an operational credential.
Active	Missing/Invalid	Reject VC-dependent service	Governance decision exists, but the node has not proven Root-issued authorization.
Inactive/Revoked	Valid	Reject new service	A historical VC cannot override latest governance state.
Missing	Valid or missing	Reject	Root cannot infer infrastructure authority without governance state.

Operational Rule

Root maintains a local authorization cache derived from the Trust Indexer and applies it before serving Registrar, Discovery, or third-party VC Issuer nodes.

Source: On-chain bulletin design; Root-indexer integration design; genesis authorization tests

Service-Node Admission Lifecycle

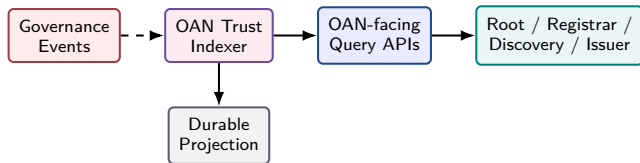


Operational Meaning

Application review, governance signaling, and Root-issued authorization are separate steps, and each step leaves evidence for audit.

Source: On-chain bulletin: Trial-network Admission Workflow; trust indexer design

Trust Indexer: Decoupling Nodes from Governance Runtime



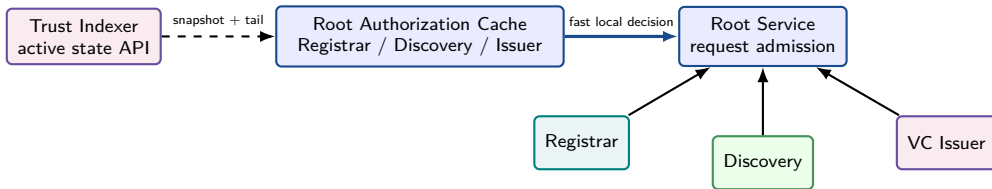
- Reads governance events.
- Reconstructs current subject state.
- Exposes stable OAN trust APIs.
- Keeps service nodes away from raw governance runtime complexity.

Design Rule

Service nodes consume OAN-oriented trust state from the indexer; the indexer absorbs event pagination, replay, projection, and cursor management.

Source: Trust indexer design: Position, Goals, Normalized Domain Model

Root Runtime Enforcement



- Root does not query raw governance runtime on every service interaction.
- It synchronizes governance state, caches active service-node authority, and stops serving revoked nodes.

Source: Root-indexer integration design; oan-reference-services Root admission logic

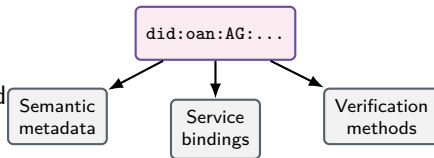
Governance State and Authorization VC Boundary

Item	What It Proves	What It Does Not Prove
Governance State	The infrastructure subject is active, suspended, or revoked under governance.	It is not a protocol credential and does not prove key control by itself.
Root-issued Authorization VC	Root issued an operational authorization credential to a DID subject.	It is not sufficient if latest governance state is inactive, missing, expired, or revoked.
Subject-Control Proof	Applicant controls the DID key used for the request.	It does not replace governance approval or Root authorization.
Signed Request Envelope	The request body, path, method, audience, nonce, and timestamp were signed.	It does not authorize an unapproved infrastructure node.

Source: Identity trust layering; on-chain bulletin: Architectural Position; oan-service-security

Why did: oan

- Portable identifier for AI-agent-related resources.
- DID document carries verification methods and service bindings.
- Semantic metadata supports discovery by user need.
- Independent of specific invocation protocols.



Design Intent

did: oan keeps the identifier stable while endpoints, semantic metadata, package versions, and protocol bindings evolve.

Source: DID method spec; did-oan refactor plan; oan-did-oan code

One DID Method, Four Product Forms

Product Form	Code	Discovery Meaning
Agent Service Skill	AG	Callable agent service with endpoint and protocol bindings.
	SK	Capability description with artifact references and implementation links.
MCP Server Tool/API	MC	MCP-compatible server exposing tools, resources, or prompts.
	TL	Callable API, connector, workflow, or tool interface.

Shared Model

All four use `did: oan`, DID document metadata, `ResourcePackage`, Root proof, and Discovery indexing.

Source: Product model: Resource Types; oan-core ResourceType subject-code mapping

Resource Forms: What Is Registered and Found

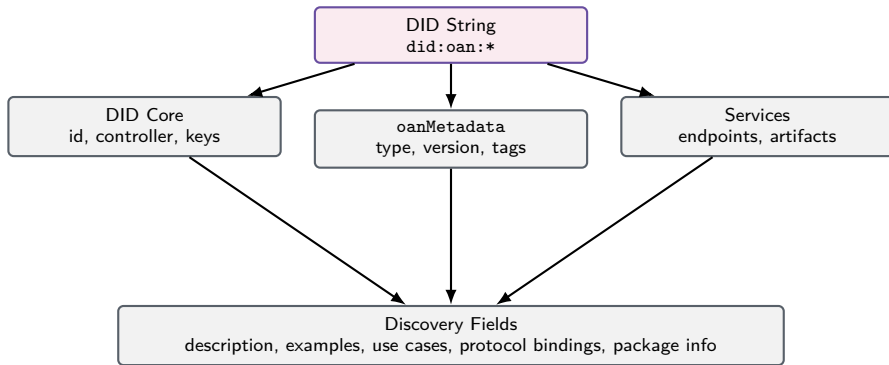
Form	Registered DID Document Carries	Discovery Returns	Consumer Next Step
Agent Service	Endpoint, protocol bindings, semantic profile, keys, VC hints.	Candidate service DID documents and signed evidence.	Verify evidence, then call via supported protocol.
Skill	Semantic description, artifact URI/hash, examples, compatibility metadata.	Candidate Skill DID documents and artifact references.	Fetch artifact from referenced location and verify hash.
MCP Server	MCP endpoint, capabilities summary, resource/tool/prompt metadata.	Candidate MCP server DID documents.	Verify, then connect using MCP.
Tool/API	Endpoint or OpenAPI reference, authentication hints, operation semantics.	Candidate API/tool DID documents.	Verify, then invoke according to API contract.

Unification Point

The resource type changes the expected metadata, but registration, Root verification, package proof, and Discovery evidence remain shared.

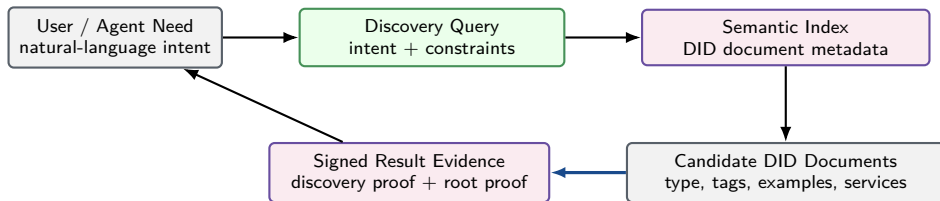
Source: Product model: four product forms; DID method spec: oanMetadata and services

DID Document as the Discovery Surface



Source: DID method spec; oan-core OanMetadata; product model Discovery Flow

Semantic Discovery Is Built on DID Documents

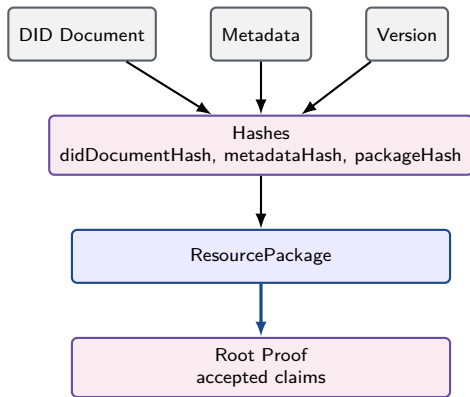


Important Boundary

Discovery indexes and returns DID documents with evidence. It does not need to host every external artifact referenced by a resource.

Source: GRAIL paper for semantic discovery direction; product model: discovery boundary

ResourcePackage and Root Proof



Source: Yellow paper: Protocol Objects; oan-package ResourcePackage and claims

ResourcePackage Binding Rules

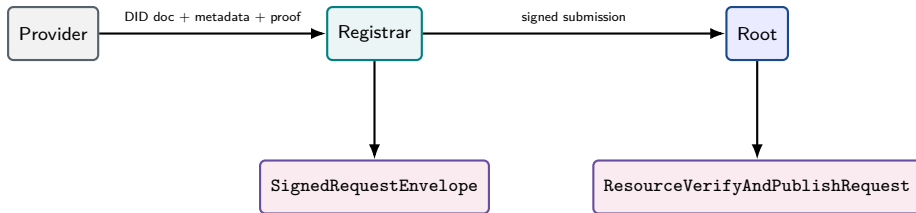
Binding	Slide-Level Meaning
<code>resourceDid</code>	Stable resource subject; not a mutable endpoint or version string.
<code>resourceType</code>	Must match DID subject code and DID document <code>oanMetadata</code> .
<code>packageVersion</code>	Versioned Root-verified release state for the resource.
<code>didDocumentHash</code>	Binds the submitted DID document to Root acceptance.
<code>metadataHash</code>	Binds semantic and product metadata.
<code>packageHash</code>	Binds the package artifact and versioned publication unit.
<code>rootProof</code>	Turns Root acceptance into verifiable downstream evidence.

Why These Bindings Matter

Consumers do not need to trust Discovery blindly; they can verify that returned documents and metadata match Root-accepted package facts.

Source: `oan-package ResourcePackage`; `did-oan maintenance guide hash/proof rules`

Resource Registration Sequence



Root verifies Registrar authorization, hashes, type consistency, freshness, nonce, and subject-control proof.

Source: System design Trust Flow; oan-protocol ResourceVerifyAndPublishRequest; registrar/root code

Publication and Synchronization



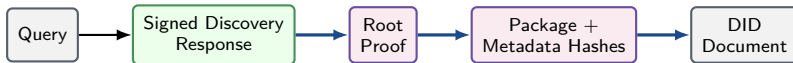
- CDN distributes Root-approved packages; it is not the trust authority.
- Discovery indexes verified packages and DID document metadata.

Pipeline Boundary

Publication may be asynchronous, but every downstream copy must remain traceable to the same Root proof, package hash, and DID document hash.

Source: System design route graph; root repository queues; CDN and Discovery service code

Trusted Discovery Chain



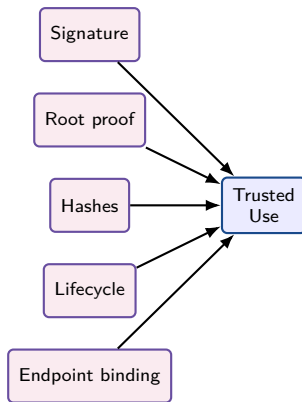
Discovery with Evidence

Discovery returns candidates that can be traced back to Root-verified resource facts.

Source: Yellow paper: Discovery Response and verification algorithms; oan-protocol ResourceDiscoveryResponse

What a Consumer Verifies

- Discovery response signature.
- Root proof and accepted claims.
- DID document hash and metadata hash.
- Package hash and hash algorithm.
- Resource type, lifecycle state, version.
- Endpoint or artifact binding.



Source: Product model: Authentication and Trust Checks; identity trust layering Runtime Verification

Discovery Query Contract

Query Inputs

- need description and keywords
- optional `resourceType`
- capability tags and custom tags
- protocol binding or endpoint preference
- version or lifecycle constraints

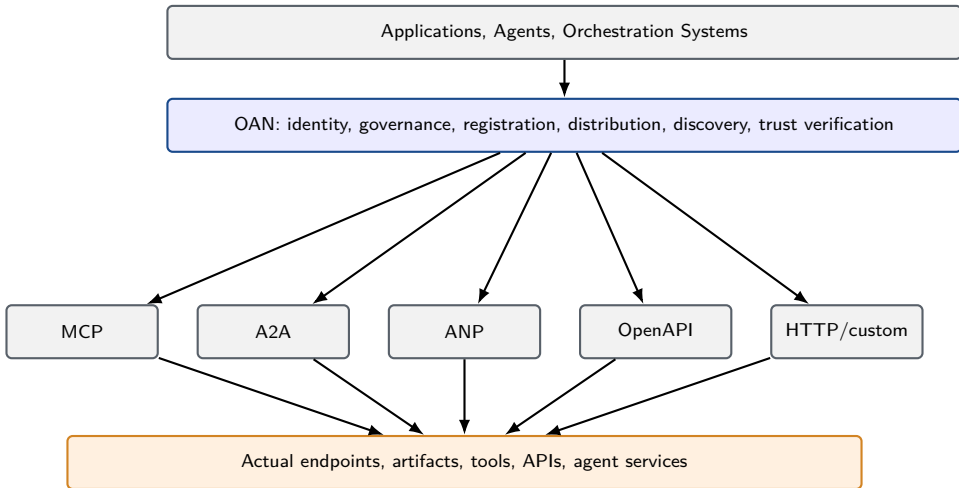
Response Evidence

- candidate `resourceDid`
- type, version, lifecycle state
- capability tags and services
- package info and Root proof reference
- Discovery response proof



Source: oan-protocol `ResourceDiscoveryQuery/Response`; `discovery-node` query implementation

OAN Above Heterogeneous Protocols



Source: White paper: relationship with MCP/A2A/ANP; yellow paper: compatibility boundaries

MCP Server and Tool/API Mapping

MCP Server

- did:oaan resource type: `mcp_server`.
- Describes endpoint, transport, protocol version.
- Summarizes tools, resources, prompts.

Tool/API

- did:oaan resource type: `tool_api`.
- Links API schema or OpenAPI description.
- Describes auth, endpoint, examples, checksum.

Source: Product model: MCP Server and Tool/API; refactor plan resource-type profiles

Agent Service and Skill Mapping

Agent Service

- Callable business Agent.
- Endpoint and protocol bindings.
- May expose A2A, MCP, HTTP, or custom protocols.

Skill

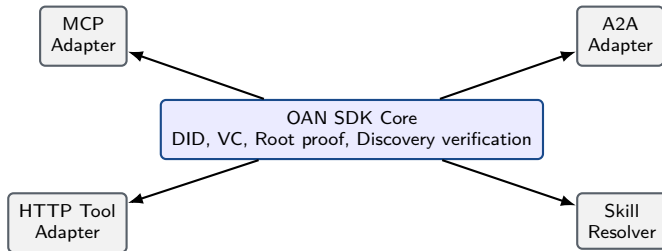
- Capability description.
- Artifact URL, hash, version.
- Implementation links to Agent Service, MCP Server, or Tool/API.

Boundary

Discovery indexes DID document metadata and references; it does not host Skill files by default.

Source: Product model: Agent Service, Skill, External Artifacts

Adapter and SDK Path

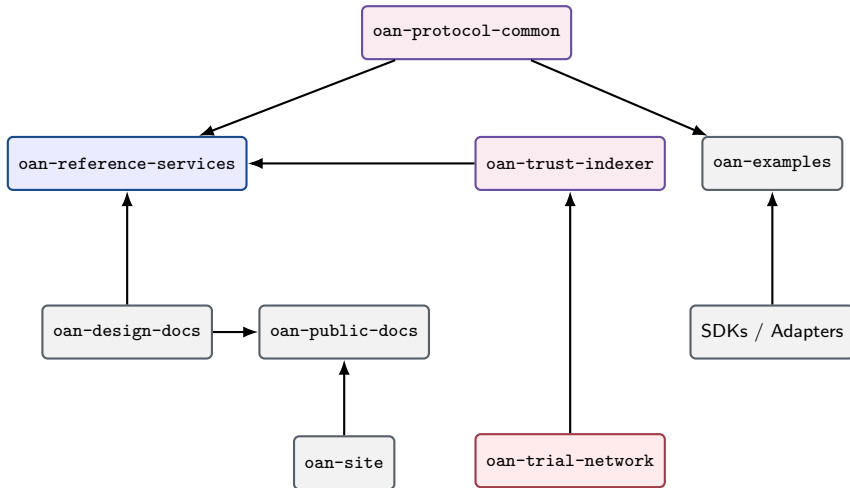


Goal

Developers should not hand-roll trust verification; SDKs and adapters should make safe use the default.

Source: Product model: Adapter SDK Direction; yellow paper: SDK Architecture

Multi-Repository Engineering Stack



Source: System design: Cross-Repository Responsibility Split

Reference Services and Shared Objects

Rust Reference Services

- Root Node
- Registrar Node
- Discovery Node
- CDN Service

Shared Protocol Crates

- oan-core, oan-did-oan
- oan-protocol, oan-package
- oan-crypto, oan-credentials
- oan-storage, oan-client

- PostgreSQL-oriented operational state; SQLite remains useful for local development and tests.

Source: System design: Implementation Stack; database abstraction analysis; protocol common crates

Current Formal API Surface

Role	Route Prefix	Purpose
Root	/root/...	Verification, publication, bulletin, authorization, queue operation.
Registrar	/registrar/..., /resources/...	Resource onboarding, registration submission, capability tree access.
Discovery	/discovery/...	Resource synchronization, query, route lookup, signed discovery responses.
CDN	/cdn/resources/..., /cdn/documents/..., /cdn/metadata/...	Distribution of Root-verified packages and document/metadata views.

Current Boundary

The current formal API surface does not retain legacy `/api/v1/...` routes in the mainline service path.

Source: System design: Formal Implementation Boundary; service main.rs route definitions

Documentation and Specification Assets

Asset	Role
White Paper	Vision, value, governance, ecosystem positioning.
Yellow Paper	Technical architecture, protocol objects, lifecycle, security.
DID Method Spec	Public specification surface for did: oan.
Detailed Design Docs	System, governance, resource model, indexer, database, website.
Public Docs	Community-facing papers, specs, and selected PDFs.
Slides	Communication material for leadership, partners, developers, and standards.

Message

OAN is being built as code + specifications + governance materials, not only as a service demo.

Source: [whitepaper](#)/[yellowpaper](#); [did-oan spec](#); [docs/system](#); [oan-public-docs](#)

Public Papers and Standards Output

White / Yellow Papers

OpenAgenet/OAN White Paper:
Open Infrastructure for Trusted Agent
Interconnection
arXiv:2606.03161

OpenAgenet/OAN Yellow Paper:
Technical Architecture for Trust-Governed
Resource Identity and Discovery
arXiv:2606.03163

Academic Paper

Existing paper
Trusted Management Model for the
Lifecycle of Agent Identity Documents

IETF Draft

`draft-xu-oan-resource-
identity-discovery-00`
Trust-Governed Resource Identity and
Discovery Architecture for
OpenAgenet

Message

OAN is moving as an open-source engineering project, a public research output, and a standards-facing architecture proposal.

Engineering Evidence Behind the Story

Claim	Evidence in Repositories	Why It Matters
Unified resource identity	oan-core, oan-did-oan, DID method spec	Identity rules are shared by services, docs, and tests.
Governed service nodes	governance design, Trust Indexer, genesis identities	Infrastructure authorization is observable and testable.
Root-verified publication	oan-package, Root service, CDN service	Discovery results can trace back to Root acceptance.
Multi-resource support	examples and protocol models for Agent, Skill, MCP, Tool/API	OAN is not limited to one Agent service form.
Developer adoption path	SDK, adapters, examples, public docs, website	Ecosystem users get practical onboarding material.

Evidence Pattern

Each major claim is backed by both design material and implementation repositories, making the project inspectable rather than merely conceptual.

Source: oan-protocol-common crates; reference services; oan-examples; oan-public-docs

Security and Trust Properties

Infrastructure Trust

- governance state is auditable;
- Root authorization VC is explicit;
- service-node requests are signed;
- revoked nodes are rejected by Root cache.

Resource Trust

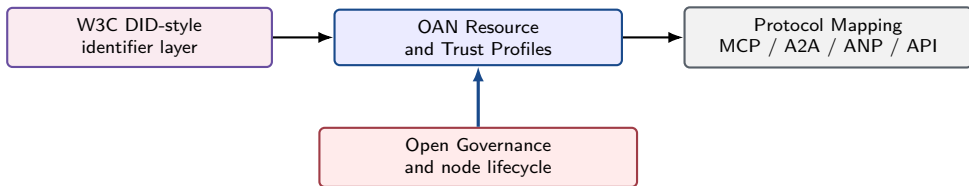
- DID document binds keys and services;
- ResourcePackage binds hashes and version;
- Root proof binds accepted claims;
- Discovery response carries verifiable evidence.

Scope

OAN strengthens identity, lifecycle, and discovery trust; it does not replace business-level safety, model evaluation, or endpoint-specific security controls.

Source: Yellow paper security model; identity trust layering; DID method spec security considerations

Standards-Facing Positioning



- OAN can be presented as a resource identity and trusted discovery profile for agent ecosystems.
- The project is suitable for open-source cooperation first, and standards-facing discussion as the model stabilizes.

Source: DID method spec; public docs; white/yellow paper standards discussion

Benchmark Baseline: Single-Node Agent Service

Topology: 1 Root + 1 Registrar + 1 Discovery + 1 CDN, benchmarking Agent Service registration and trusted discovery under the default PostgreSQL path.

Scale	Reg.	Disc.	Miss.	Reg TPS	E2E TPS	Total ms	Submit P50	Submit P95	Visibility P95	Query P95
10	10	10	0	8.02	4.94	2023.77	669.19	1226.33	2286	13.57
50	50	50	0	35.26	17.28	2892.82	74.57	1187.73	2942	46.69
100	100	100	0	37.00	20.06	4980.00	74.00	2090.00	4880	73.00
500	500	500	0	41.46	31.02	16120.00	553.00	2744.00	12006	442.00
1000	1000	1000	0	23.88	22.58	44280.00	937.00	1663.00	12363	884.00

Source: oan-examples benchmark reports: pipeline-e2e PostgreSQL current baseline

Benchmark Baseline: Single-Node Mixed Resources

Topology: 1 Root + 1 Registrar + 1 Discovery + 1 CDN, benchmarking mixed Agent Service / Skill / MCP Server / Tool API publication and trusted discovery.

Scale	Reg.	Disc.	Miss.	Reg TPS	E2E TPS	Total ms	Submit P50	Submit P95	Visibility P95	Query P95
40	40	40	0	16.60	10.23	3910.00	82.00	2017.00	3678	84.00
100	100	100	0	52.42	26.98	3710.00	99.00	1562.00	3722	113.00
200	200	200	0	45.04	32.52	6150.00	183.00	937.00	4353	184.00
500	500	500	0	34.20	27.90	17920.00	882.00	1356.00	11870	439.00
1000	1000	1000	0	19.50	18.78	53250.00	1177.00	1897.00	13910	796.00

Source: oan-examples benchmark reports: mixed-resource-pipeline-e2e PostgreSQL current baseline

Benchmark Baseline: Multi-Node Agent Service

*Topology: 1 Root +
3 Registrars + 2 Discoveries + 1 CDN, with the scale shown below interpreted as per-Registrar registration volume.*

Per-Reg.	Reg.	Disc.	Miss.	Reg TPS	E2E TPS	Total ms	Submit P50	Submit P95	Visibility P95	Query P95
10	30	30	0	9.76	8.37	3580.00	1257.00	1523.00	2651	43.00
50	150	150	0	24.97	23.60	6360.00	105.00	3773.00	5131	100.00
100	300	300	0	39.12	31.98	9380.00	241.00	2669.00	6822	188.00
200	600	600	0	28.50	24.09	24910.00	970.00	1508.00	14321	637.00
400	1200	1200	0	17.19	15.37	78060.00	2045.00	2920.00	42598	1234.00

Source: oan-examples benchmark reports: multi-node-pipeline-e2e PostgreSQL current baseline

Benchmark Baseline: Multi-Node Mixed Resources

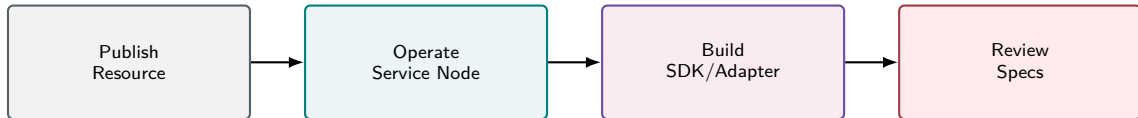
Topology: 1 Root + 3 Registrars + 2 Discoveries + 1

CDN, benchmarking mixed resource publication and trusted discovery with total scale distributed across Registrars.

Total Scale	Reg.	Disc.	Miss.	Reg TPS	E2E TPS	Total ms	Submit P50	Submit P95	Visibility P95	Query P95
60	60	60	0	16.48	13.20	4540.00	762.00	1594.00	4049	96.00
150	150	150	0	34.00	18.00	8330.00	76.00	2591.00	7926	164.00
300	300	300	0	53.03	28.63	10480.00	528.00	866.00	7004	278.00
600	600	600	0	27.28	21.74	27600.00	1262.00	1968.00	19571	517.00
1000	1000	1000	0	16.61	14.99	66690.00	2107.00	2967.00	42311	818.00

Source: oan-examples benchmark reports: multi-node-mixed-resource-pipeline-e2e PostgreSQL current baseline

How to Participate



- Publish Agent Service, Skill, MCP Server, or Tool/API resources.
- Apply to operate Registrar, Discovery, or future VC Issuer nodes.
- Contribute adapters, examples, documentation, and conformance tests.

Source: On-chain bulletin admission workflow; product model Registration/Discovery Flow; oan-examples

Short Term

- Public docs and slides.
- DID method refinement.
- SDK/adapters.
- Trial network material.

Medium Term

- Partner-operated nodes.
- Conformance tests.
- Protocol mappings.
- More examples.

Long Term

- Federated trust domains.
- Cross-domain discovery.
- Community governance.
- Standards work.

Source: White paper Roadmap; yellow paper Open Technical Work; current design docs



OAN builds open governance infrastructure for trusted agent interconnection.

Publish
resources

Operate
service nodes

Build
SDKs and adapters

Review
specifications

<https://github.com/OpenAgenet>

xujinliang@caict.ac.cn; jlxfly@gmail.com

Source: White paper conclusion and stakeholder value; public docs